

LINEAR PROGRAMMING (LPP)

MCQ SHORTCUT TRICKS — BY THUNDERSTUDY

TRICK 1 — Corner Point Rule (The Golden Law)

The optimal solution (Max or Min) **ALWAYS** lives at a corner point (vertex).

MCQ Hack: Don't draw the graph. Substitute the corner points from options directly into Z!

TRICK 2 — Inequality & Shading (2-Sec Rule)

- $\leq$ (Less than or equal): Shade Towards origin (0,0).
- $\geq$ (Greater than or equal): Shade Away from origin.

Quick Check: Put (0,0) into the equation. If it works, origin is included!

TRICK 3 — Infinite vs No Solution

Infinite Solutions: If the Objective Function (Z) has the same coefficients as a constraint (parallel lines).

No Solution: If constraints are opposite (e.g., $x+y \geq 10$ and $x+y \leq 2$). Impossibility = No region.

TRICK 4 — Fast Option Elimination

1. If $x, y \geq 0$, immediately reject any option with negative numbers 
2. Test if the point satisfies all constraints. If it fails even one, reject it!
3. Substitute remaining valid points into **Z**. Choose the winner.



TRICK 5 — Quick Substitution Example

Maximize: $Z = 3x + 2y$ | Points: $(0,0)$, $(2,0)$, $(1,3)$, $(3,1)$

Point	Calculation	Result
$(0,0)$	$3(0) + 2(0)$	0
$(2,0)$	$3(2) + 2(0)$	6
$(1,3)$	$3(1) + 2(3)$	9
$(3,1)$	$3(3) + 2(1)$	11 (Winner)



BOARD EXAM SPEED STRATEGY

"Skip the graph, test the points, find the corners."

- 1. Read Z first.
- 2. Check provided vertices in options.
- 3. Validate against constraints.
- 4. Pick Max/Min.

THUNDERSTUDY: NO GRAPH, NO STRESS